

From Point A to Point B: Identity-Informed Safety in Navigation Technology

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ABSTRACT

HCI scholars have explored how safety and mobility intersect in urban environments, particularly for women and people with disabilities. However, existing work has yet to explore how intersecting identities shape perceptions of safety and navigation strategies. To better understand these relationships, we conducted 12 semi-structured interviews with a diverse group of Seattle residents who regularly navigate the city. Age, gender, race, ability, and addiction recovery shaped how participants characterized a route's safety and their navigation practices. Technology served multiple roles in safe navigation — some in tension — by surfacing route information, distracting from surroundings, and empowering decision-making. Our findings highlight opportunities for future research and navigational technology design to meet the needs of differing safety experiences by incorporating personalization, relevant data, and guidance.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Empirical studies in interaction design;

KEYWORDS

Navigation technology, mapping applications, identity, race, gender identity, intersectionality, safety, urban navigation

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1 INTRODUCTION

Mobile navigation applications can empower individuals to traverse unfamiliar and ever-changing urban landscapes shaped by longer-term transportation and urban policy debates. However, urban navigation using modes of transportation other than driving — such as walking, biking, and riding public transportation — adds unique challenges to navigation and personal safety, particularly for minoritized individuals [1–7, 13]. A growing area of HCI is exploring the expansion of navigation tools to support mobility priorities such as speed, convenience, fuel efficiency, and safety; for example, prior work on pedestrian navigation safety has focused

on women's experiences of sexual harassment [1–7] and issues of accessibility on urban streets [8–11], particularly for communities in Southeast and East Asia [1–4]. However, prior work has also highlighted that experiences and conceptions of safety are highly contextual based on location and personal identity [4, 6, 12–14].

To deepen understanding of safety in urban navigation, this study explored how people with diverse identities engage with technology to negotiate safety while navigating urban spaces in Seattle, Washington, US. Through 12 semi-structured interviews, we found that compounding age, gender, race, ability, and addiction recovery status shaped how participants characterized a route's safety and their navigational practices. We found tensions in how navigational technologies both contributed to and detracted from safe navigation by providing safety-related route information, empowering decision-making, yet distracting from surroundings. Our research can help to inform the design of navigational technologies that better support situated safety.

2 BACKGROUND

2.1 Intersectional Safety & Navigation Technology

HCI scholars have advanced intersectionality [14–19] as a critical lens for designing technologies that are responsive to the nuances of identity formation and performance in a range of settings. In parallel, recent work has explored the role of mobile technologies in navigating urban environments safely, particularly for women, transfolk, and people with disabilities. In studying middle-class women in Bangalore and San Francisco, Blom et al. [1] found that darkness, the potential for transit accidents or theft, men displaying concerning behavior (i.e., intoxication), and unfamiliar men appearing to be of lower socioeconomic status made pedestrian settings feel unsafe for women. Mobile technologies played an important role in women feeling safer through protective tactics like pretending to use their phones, calling or texting friends, taking photos of the perceived threat, and changing their routes. The authors tested a prototype to capture and share location-based safety attributes like distance to the nearest authority, proximity to members of one's social network, the illumination level of streets, the presence of familiar landmarks, and crime rates. However, the authors found that sharing location data with other app users did not make them feel safer.

Additional work has explored gender harassment as a safety concern for women in urban settings and reinforces Blom et al.'s assertion that navigation algorithms can lighten the burden on women to review and plan safe routes. For instance, Yoo and Dourish [2] explored a mobile app called Anshimi, created to provide navigation support and easy access to police reporting of street harassment. The authors found that women observe multiple factors in determining a location's safety, including time of day, amount

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of light, transportation method, type of road, and nearby crowds. However, the authors found that any safety algorithms must be transparent so women can trust suggested route guidance and that GPS-based mobile applications can cause women to pay attention to their phones instead of their surroundings. Similarly, Ahmed et al. [3] and Ali et al. [4] both investigated mobile solutions to sexual harassment in Bangladesh. Ahmed et al. developed an application, Protobadi, to report, map, and share women's stories about sexual harassment. The authors found that using data to highlight the regional prevalence of sexual harassment altered women's behavior rather than that of potential attackers, thus making those areas even more dangerous for women who unknowingly pass through them. Ali et al. observe that sexual harassment is "spatiotemporal in nature, that is, a certain type of harassment occurs more frequently at a certain place at a particular time." They designed a mobile application called SafeStreet that crowdsources women's experiences of street harassment, visualizes aggregated reports on a map, and uses harassment data to determine the safest time of day and path for users to reach their destination. Similarly, HarassMap, another application mapping sexual harassment in Egypt, leveraged crowdsourced data to advance community mobilization campaigns offline [5].

Satchell and Foth [6] explored how men and women balance privacy concerns with feelings of fear and vulnerability in urban areas after dark. The authors found that both men and women (there were no non-binary participants) voiced safety concerns depending on location, transportation modality, and the presence of others, but that men took fewer precautions using mobile technology than women. The authors advocate for designing technologies to enhance the enjoyment of the city instead of paranoia [7].

Non-binary people, transfolk, and people of color experience compounding concerns around safety, including racialized and gendered violence. In online communities, Scheuerman et al. [12] observed that non-binary people, particularly non-binary people of color, experience targeted forms of harassment and violence. The authors call for tools that empower these groups to connect and build community without the fear of inviting more digitally enabled harassment. Starks et al. [13] similarly explore creating digital tools to empower transfolk and nonbinary people of color to exist safely. The authors found that participants had to navigate not only physical violence from police and others in public spaces but also psychological violence from transphobic misgendering and harassment. Through a tool called U-Signal, they aimed to support safe navigation by checking in with trusted contacts and building community with other transfolk.

In Seattle, US, Woelfler et al. [20] used value-sensitive design [21, 22] to improve safety among homeless youth. Using open-ended methods that allowed the participants to elicit personal perceptions and experiences with the term "safety," the authors found that safety is situational for homeless youth, who often have shifting levels of trust among peers, police, and social workers. The authors found a need to document abuse, reach out for help without being visible, and support access to basic infrastructure like electricity for charging without entering areas where they were surveilled and stigmatized. This work highlights the intersections of place, mobile technology, and safety for a historically underrepresented group.

2.2 Technical approaches to safe and accessible navigation

Technical solutions for safer pedestrian navigation regularly use publicly available datasets [23–25] or crowdsourced data [26, 27] to surface safer routing options. Beyond visualizing sites of violence and harassment, technical approaches to support safe urban navigation have foregrounded crowdsourced tracking of mobility barriers [8, 9], gamification [8], lighting support [28–31], and personalization [32], particularly for people with disabilities and unsheltered individuals. For example, the Safe Street Rangers project [33] in Thailand used crowdsourced data to report safety factors like street damage, road obstacles, broken traffic signs, risky turns, unattended dogs, dark streets, and unfamiliar people, thus enabling users to plan safe pedestrian routes based on reported issues. Saha et al. [8] also use crowdsourcing to overcome the challenge of mapping street-level accessibility data through a large-scale digital tool called Open Sidewalk, which uses street-view technology to enable users to virtually traverse a physical space and denote potential accessibility issues. The authors propose gamification to encourage users to log this accessibility data using missions to encourage real-world exploration documentation of accessibility issues.

Building on crowdsourcing approaches, Cena et al. [10] explore a map personalization tool for people with Autism Spectrum Disorder (ASD) to navigate to "safe points of interest" depending on their preferences and aversions to sensory characteristics. The authors proposed using crowdsourced data to build a knowledge base of personalized recommendations for ASD users. In another study exploring cognitive decline and navigation, Rassmus-Gröhn and Magnusson [11] find that turn-by-turn instructions are not ideal for older adults experiencing memory issues and tested two mobile app prototypes, one using compass navigation and the other utilizing landmark-based wayfinding; both were effective at helping users with memory problems navigate home.

2.3 Research gaps and opportunities

Together, these papers highlight the geographic and cultural specificity of safety concepts for people with diverse identities, as well as the tensions with privacy and trusted information in designing to support them. They also highlight the breadth of interaction techniques that support safe navigation, from improving lighting to highlighting crime statistics to crowdsourcing accessibility and harassment concerns. However, existing work around urban safety concerns largely focuses on single-axis differences in identity, a focus that Schlesinger et al. [15] and Wong-Villacres et al. [16] (building on the foundational work of Black feminist legal scholar Kimberlé Crenshaw [14] and postcolonial feminist political scientist Rita Kaur Daumoon [17, 18], respectively) argue is limiting to understanding how multiple compounding identities [14] play out situationally.

In particular, the role of race in safely navigating urban spaces has been largely neglected in HCI scholarship thus far, a significant gap in the context of racially motivated violence that continues to impact people of color disproportionately. We take up the calls of Ogbonnaya-Ogburu et al. [34], Schlesinger et al. [15], and Wong-Villacres et al. [16] to amplify stories from people of color, disclose

our own positionality, and report nuances of context and subjectivity in approaching this study of safety concerns while navigating urban spaces. Beyond single-axis analysis of binary gender and disability status, we investigate how individuals with compounding racial identities, socioeconomic status, mobility, age, and addiction recovery status understand and navigate safety in urban spaces with in-depth interviews with a diverse group of respondents in Seattle, WA, US. Specifically, we asked: How do people with diverse identities perceive and define safety while navigating urban spaces as pedestrians? How do individuals' identities and embodied experiences impact how they use technology to navigate urban spaces and conceptualize safety?

3 METHODS

To answer these questions we used semi-structured interviewing [35, 36] due to its effectiveness in eliciting perceptions, stories, and lived experiences of individuals with diverse identities [34]. The authors' university Institutional Review Board [STUDY00014798] reviewed the ethics of the study before recruitment began. We scoped our research geographically to residents of Seattle, WA, and recruited 12 participants from multiple neighborhoods. All participants reported walking weekly as a primary form of navigation and using mobile apps for directions. Eight women, three men, and one non-binary person participated in the study, and we recruited more women than men given the prevalence of street harassment women experience [43]. We deliberately recruited people of color to learn how race influences pedestrian safety; participant demographics can be found in appendix 1. Interviews lasted 30–60 minutes and were conducted over Zoom video during February and March 2022; remote interviews were chosen over in-situ methods to ensure participants and researchers were not exposed to COVID-19 infection. All conversations were recorded with consent.

We conducted a thematic analysis followed by affinity mapping [35–37] to identify common and divergent themes. We used a digital whiteboard to inductively code individual stories and quotes, then clustered similar findings into themes across participants, annotating divergences and tensions alongside themes. In total, we identified more than 50 themes from the data covering topics like identity, navigation strategies, and usage of navigation technologies. We clustered themes that were related to each other and narrowed down to memo what we present here based on their relevance to identity-informed safety and its impact on navigation technology.

Our methods allowed us to elicit rich stories and perceptions around safe urban navigation and technology, yet also had specific limitations. Firstly, remote interviews allowed us to safely interact with a diverse group of participants while the COVID-19 pandemic was still a prominent safety concern; however, it limited our ability to gather in-situ data while participants navigated physical spaces. Secondly, we intentionally recruited a broad group of participants in terms of age, gender, race, and neighborhood, to capture a wide cross-section of perspectives. However, this is a small sample size, so these findings are intended to amplify stories [34] and describe the complexities of identity-driven navigation; they are not necessarily generalizable to the experiences of entire populations.

To better situate the researchers' approach to this work, two members of our team identify as white women, and one identifies

as an Asian and white man. All team members are between the ages of 30 and 40 with no disabilities that impact mobility, have stable housing, and have, or are pursuing, advanced degrees in human-computer interaction. Two members of our research team also work professionally in the tech industry. We recognize that our genders, races, ages, mobility, and socio-economic security influence our approach to this inquiry and interpretations of the information shared during interviews.

4 FINDINGS

4.1 Defining Safety

Intersecting aspects of participants' identities directly shaped their definitions and perceptions of safety. Although participants mentioned multiple identities during our interviews, age, gender, ability, race, addiction recovery status, and housing status emerged prominently as influences on how they conceptualized safety while navigating urban spaces.

For older participants (those over 65), perceptions of a route's safety were shaped by the risk of falls causing injury. Uneven sidewalks, for example, were mentioned by all three elder participants as a safety risk, with one participant describing safe navigation as "smooth paths with no tripping hazards" [P3: woman, white, age 65+]. Another participant mentioned that she avoids walking at night whenever possible, saying, "I don't like to walk too much in the dark, and that isn't because I'm afraid of crime or anything. It's because I can't see the sidewalks" [P5: woman, white, age 65+]. Participants also mentioned steep hills as an environmental consideration and described adjusting their travel to find routes with fewer steps, gradual inclines, and elevators. One participant said, "Nowadays, it's making sure I can navigate around physical limitations" [P5].

While physical limitations were central to older women's perceptions of safety, they were also very aware of how others might perceive them due to age. One participant stated, "I'm a woman, a little old woman, and I'm kind of small. So I just, sometimes, [in] environments I don't feel comfortable, I like to change that environment" [P1: woman, Irish, age 65+]. Another participant described being "more vulnerable" because of her age, potentially making her a target for attack or robbery [P3]. Although advanced age was a potential source of vulnerability, one participant also felt less at risk of harassment due to her age, stating, "I don't get catcalls. I don't get men trying to brush up against me on a bus, which are all things that, when you're a younger woman, those things definitely happen" [P5]. In these stories, we saw that advanced age intersects with both mobility and gender and can be both a source of vulnerability and security, depending on the situation.

Participants living with physical disabilities spoke about physical limitations impacting their views on safety. One participant, for example, described living with chronic pain due to fibromyalgia, which made walking difficult [P11: non-binary, Asian]. As with older participants, this person said routes with minimal hills and well-maintained sidewalks without trash were crucial characteristics of a safe route. Another participant spoke about an inner ear condition that prevented her from using escalators and described a time when the local metro app directed her to get on a train at an

underground station [P1]. In this situation, the app did not mention that the station was underground and presented no alternative routes (even though ones existed), thus impacting her ability to reach her destination safely.

The compounding safety concerns of gendered violence were a common theme. Female participants felt vulnerable traversing public spaces alone and characterized a route's safety by a risk of being attacked. One participant said, "As a woman of any age, we're just conditioned to have worries about walking alone and, like I said, perceived vulnerability" [P3]. As elder participants highlighted, physical size or "petiteness" was identified by multiple women as a source of vulnerability. As one participant put it, "Being a female, especially a petite one, that is something I always try to keep in mind. I am not going to go toe-to-toe with strangers, so I try not to even get in those situations. Which is why I choose the method of plan B or just reroute instead of walking through it" [P7: woman, Black or African American]. As this participant alluded to, the threat of violent encounters requires women to change their route or behavior to avoid the risk of being physically overpowered. Similarly, female participants expressed a need to constantly be aware of their surroundings and alert for potential danger in public spaces: "Being a woman, you're taught to be extra vigilant" [P9: woman, Asian].

The non-binary interviewee also described their risk of sexual harassment while walking. "Random people out on the streets will come up and call me baby or whatever," they said. They also received unwanted sexual advances on city buses, saying, "I see a lot of people, I guess, flirting with me on the bus too. So I just don't engage in that" [P11]. To deter these advances, the participant would often wear headphones or say, "Sorry, I'm taken," hoping to end the encounter [P11].

Men did not mention gender as a personal safety concern; one male participant stated directly: "I don't think I've ever thought about either gender or race when walking" [P2: man, Asian]. These participants were more concerned about the safety of those around them. One participant, for example, expressed that although he was not often concerned for his safety, he was hyper-aware of danger when walking with his roommate with a physical disability who was attacked multiple times [P4: man, American Indian or Alaska Native & white]. A female participant specifically recounted conversations with her male partner, who said he would cross the street to avoid making a woman feel unsafe in public [P12: woman, Asian].

Race also shaped perceptions of safety, particularly among people of color. Minoritized participants highlighted that recent US events related to anti-Asian hate and nationwide protests following the murder of George Floyd (a Black man killed by police) caused them to become more aware of their safety in public and re-evaluate their navigation. For example, one Asian woman said, "As someone who's an ethnic minority...and especially in the last couple of years where there have been a few racial hate crimes towards certain groups of the population, I think that's something that I was concerned [about]...So maybe that part of my identity comes out a little bit into the decision-making" [P12]. Another Asian participant said she had never really thought about her race until the summer of 2020 when the Proud Boys, a neo-Fascist organization, showed up in Seattle during protests following George Floyd's murder.

Following that event, she said her husband always accompanied her for safety reasons when they walked in public [P6: woman, Asian]. Anti-Asian hate was a frequent concern, with another participant describing a situation where they encountered an unhoused person with apparent mental illness yelling anti-Asian slurs at them while on a city bus [P11]. As a result, this participant associated Seattle's unhoused population with a risk of encountering anti-Asian hate, making areas with communities of unhoused people feel unsafe. "Being Asian," they said, "It's very dangerous for me to travel" [P11]. Race also influenced how participants of color engaged with people around them while navigating. For example, one participant who identified as a Black man said, "Especially when I'm walking into a neighborhood that's not busy, I'm like, 'At this time of night and I'm walking, how will they think of me? Will they see me as a danger?'" [P8: man, Black or African American]; he was acutely aware that people in his environment might perceive him as a risk due to his race. His concern impacted his ability to ask for help when navigating. He said, "It's also difficult asking people for help because of my color. The person is moving faster because he thinks, 'Maybe he's dangerous. What are you coming to ask from me?'" [P8].

Two participants stated that their recovery from drug addiction directly shaped their perceptions of safety while navigating; actively avoiding areas where they might encounter relapse triggers or people from their past using drugs was important in choosing a safe route. One participant said, "Everything I do right now is recovery-focused" [P4]. These individuals expressed that avoiding triggers was becoming more difficult due to the increasing prevalence of people openly using drugs along major transit corridors. For example, one participant said, "Downtown, Third Avenue, is not the most pleasant place, especially for someone who is new into sobriety and recovery. The amount of, you know, drugs and stuff like that...it makes it kind of challenging to want to walk in that area anymore" [P4]. The other participant recovering from addiction echoed this sentiment, saying, "Right now, it's like, there's a lot of people using outside, and so that triggers me" [P11]. These participants would avoid these major corridors when possible by identifying alternative routes to their destinations; however, these encounters were often unavoidable because they are visible along mass transit locations. Traveling through areas where one might encounter drug use required mental priming to prepare for the emotional distress the route could cause.

Across every interview, stories about safety opened discussions about Seattle's unhoused population and the perceived risk associated with navigating through these highly stigmatized communities. However, we did not actively recruit participants from unsheltered communities to shed light on their safety concerns, and how housing status intersects with race, gender, ability, and age in their safety experiences. Here, we amplify existing work that has shown limited supportive housing, regular encampment removal, and highly surveilled access to public facilities like parks and community centers constrain basic access to health and hygiene resources [38]. These conditions leave unhoused individuals vulnerable to safety concerns like targeted harassment, violence, and exposure [21].

4.2 Safety Strategies & Technology's Role

Safety practices and navigation technology often converged during route planning. Participants shared several techniques they use to determine the safest route before setting out on a journey, particularly using navigation applications to gather information to avoid unsafe areas. Most participants shared opinions about particular areas of the city that they perceived as unsafe — as specific as cross streets like 3rd and Pike and as broad as the Downtown neighborhood — and noted that they would use apps to plan routes that avoided navigating through these areas [P1, P2, P3, P4, P5, P6, P7, P9, P12]. While some participants avoided walking through these parts of town, others avoided public transit routes that required transfers in these areas as well. Still, others noted feeling unsafe while riding public transportation through unsafe parts of town. Navigation applications enabled these participants to check if a planned route intersected an area they perceived as unsafe and look for alternative routes.

Participants' motivations for categorizing a particular street or area of the city as unsafe were tied closely to their identities. As previously mentioned, participants recovering from addiction [P4, P11] avoided certain areas where they were likely to encounter drug or alcohol activity. For other participants, race and gender played a role as participants avoided areas where they previously experienced harassment or expected to experience harassment. For example, one female participant noted avoiding 3rd Avenue between Stewart Street and Pike Place due to "rough, sketchy, aggressive people" [P5].

Besides prior experience, participants also shared characteristics they use to determine whether a street is safe to navigate. Some participants preferred walking on main streets due to the presence of other people, lighting, and open businesses [P2, P3, P7, P9]. One participant was unhappy that navigation apps like Google Maps often suggest direct routes away from main streets [P2], while another had mixed feelings about main streets, noting that they would prefer to walk on smaller streets due to less people, traffic, noise, and trash [P3]; however, this person also noted purposefully taking main streets when navigating through unfamiliar neighborhoods to prioritize safety. Several participants said the number of people in an area was a key indicator of safety [P2, P3, P4, P7, P9, P11], but there was no consensus on whether more or fewer people made a space safer.

When taking public transportation, some participants — all women or non-binary — shared that they prioritize fast connections and transfers [P9, P10, P11, P12] partially to minimize time spent in unsafe environments. One participant noted that she chooses bus routes with as little walking distance as possible, especially at night [P9]. Another said finding bus routes that run near her neighborhood can be difficult and increases her feelings of vulnerability: "A lot of buses coming from Queen Anne, they drop you off at like Third Avenue, and I remember the first time that happened, it wasn't necessarily the best" [P12]. These participants used navigation applications to determine the shortest walking distance and time when choosing between multiple public transportation routes. A select few — all women — were heavy users of 360-degree, navigable images like 'Street View' in Google Maps or 'Look Around' in Apple Maps [P5, P7, P9]. These participants utilized these features

to better understand certain areas of town, streets, intersections, or bus stops before starting their journeys. Some looked for indicators of safe routes, like busy streets, while others used the view to familiarize themselves with landmarks. One participant also utilized the satellite view to find routes through residential neighborhoods and avoid streets with large communities of unhoused individuals [P7].

Participants also perceived directions from navigation applications as a safety risk. One woman noted that she keeps the audio direction turned off on navigation applications because she believed the sound from a phone made her a target [P7]. Additionally, she described only taking quick glances at the map to avoid being distracted or perceived as lost by strangers. For older participants, avoiding the distraction of technology mid-route was influenced by the desire to avoid physical injury from tripping and falling while walking [P1, P3, P5]. Although all older participants used navigation applications, they often used them to pre-plan routes instead of for turn-by-turn directions while walking to avoid distractions [P1, P3, P5]. One participant stated, "I have to make sure that I'm really concentrating on walking and looking at my surroundings and how the pavement is," [P1] while another participant said, "When you're younger, you can multitask better," [P5] implying that younger people are better able to walk and use their smartphones simultaneously. In these cases, the participants were cognizant of navigation applications' impact on their situational awareness and limited their smartphone use to avoid safety risks.

Despite these distractions, participants said navigation applications empower them to feel safer. One participant noted that map applications help him feel safer by ensuring he knows exactly where he is at any given time [P3]. Multiple participants expressed that map applications reassure them that they can find a familiar place (e.g., grocery stores, coffee shops) if they find themselves in an unsafe situation [P3, P4, P7]. These apps also empowered participants with the flexibility to plan and adapt. One person said, "One of the things that really makes me feel safer is a plan. A plan and the ability to change that plan" [P1]. Some participants viewed navigation applications as a replacement for a human travel companion. Multiple women expressed that they thought about safety less when traveling with a partner and that navigation applications served a similar purpose when traveling solo [P7, P12], noting that navigation apps act "as a companion, like traveling with a buddy" [P12] and that "even when I'm lost, I still feel like it's a physical manifestation. Well, Google's not a person, but it feels like somebody else is there, and I'm not completely alone" [P7].

5 DISCUSSION: INCORPORATING SITUATED SAFETY DATA & REMOVING DISTRACTIONS

Participants' stories in this study highlight a significant opportunity to tailor navigation technologies around situated, identity-informed notions of pedestrian safety while longer-term advocacy and policy responses unfold. Navigation applications like Google Maps, Apple Maps, and Waze currently emphasize time efficiency (i.e., getting from point A to point B as quickly as possible) in route selection with secondary options like avoidance of tolls (financial values) and fuel efficiency (sustainability values) [39]. Our findings highlight the opportunity to surface and proactively share

more safety data within these applications to inform situated and identity-informed decision-making. For example, customizable user preferences could adjust map layers to display public safety hubs (e.g., grocery stores and coffee shops) and customize routes based on individuals' safety preferences (e.g., preferences for well-lit streets or main roads, avoiding alleys). Specific participant suggestions included data on the presence or absence of sidewalks [P10: woman, Asian], real-time crowdsourced data about bus route conditions [P7], and geographic data highlighting drug-related activity [P4]. Many navigation apps offer alerts for road closures, increased traffic, and the presence of police vehicles, but comparable pedestrian-focused alerts are lacking.

Although the identities explored in this study transcend the boundaries of a single city, how they intersect with safety depends on local social, economic, and political conditions. This variability raises the question of how to incorporate situated, identity-informed safety data into navigation apps while bridging the sociotechnical gap [44] between emergent, technologically entangled practices and app functionality. One possible solution is through crowdsourcing. Building upon past studies exploring crowdsourced sexual harassment [2–5], street lighting [28, 30], and sidewalk accessibility data [8, 9], future research can examine ways to incorporate crowdsourced data within major navigation systems to address the breadth of safety needs beyond Seattle. Although promising, multiple studies have highlighted the limitations of crowdsourcing such as sampling bias [40], under-reporting in certain locations [41], and challenges in motivating users to submit data [42]. Thus, any crowdsourced solutions would need to find ways to address these challenges to ensure diverse safety needs are accurately represented in map data.

Allowing access to ambient route directions and contextual geographic information with minimal interruption to attention might empower pedestrians to utilize navigation apps without the risk of distraction. Participants also expressed that inaccurate or unclear directions can pose a safety risk as additional attention is required to determine the correct route [P6, P7, P8, P12]. Here again, crowd-sourcing data could empower users to report inaccuracies and decrease the safety risk of distracting wrong directions.

For some participants, navigation apps acted as trusted companions that empowered them to navigate the city safely. As one woman said, “By myself, I would probably not venture out at all in Seattle. . . But because of Google [Maps], I have the power and, like, the mental fortitude to be able to try these things on my own” [P7]. Supporting this sense of empowerment through companionship is a critical design opportunity that has been under-explored in past research on navigation technology. For example, one participant suggested enabling users to send their live location to a trusted friend or family member and automatically alerting them if they depart from an expected route or identify potential safety risks [P12].

This study also highlights the value of an intersectional lens on safety. Our findings extend prior HCI research on women's safety [1–5, 7] by revealing how gender intersects with race and age to shape perceptions of pedestrian safety. For example, concerns about sexual harassment were often compounded by concerns of racial violence for women of color. Similarly, we extend existing

literature on accessible navigation [8, 11, 33] by discussing how sidewalk accessibility issues are a critical safety challenge for women over 65 and are often more concerning than fears of sexual harassment. This study also revealed unique facets of identity that past literature has largely overlooked. For example, addiction recovery was a topic we did not set out to examine, but that emerged as a significant safety consideration shaping navigation practices for two participants. Similarly, past research largely overlooks how race and other identities intersect to shape technology-assisted pedestrian navigation. However, this study shows that intersections between identities are an important influence on technology's role in navigating from point A to point B.

6 CONCLUSION

Mobile navigation applications have transformed how individuals traverse urban spaces. This research explored how individuals perceive safety while navigating urban environments and how technology shapes those perceptions of safety. Age, gender, race, physical ability, and recovery from addiction influenced how participants evaluated the safety of a route and altered their behavior to protect themselves while navigating. Navigation apps, such as Google Maps and regional transit apps, were utilized to determine an efficient route and find alternatives when a route was deemed unsafe. These applications also empowered some individuals by providing a source of independence and comfort while traversing unfamiliar environments. Conversely, participants also noted ways navigation apps fall short and detract from their feelings of safety. This study points to multiple opportunities to improve the design of navigation apps to meet diverse safety needs. Some key opportunities include supplementing information about travel efficiency with information about a route's safety, enabling personalization of navigation apps to meet varying definitions of safety, limiting the distraction created by navigation apps while walking, and finding new opportunities to foster a sense of empowerment. Multiple opportunities exist to build upon this and other research exploring the intersection between identity, technology, and navigation safety. Potential areas for future exploration include supporting the safety needs of unhoused individuals, expanding the geographic reach of this study to other settings, and further exploring the use of crowdsourced data to augment safety information in navigation apps.

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REFERENCES

- [1] Jan Blom, Divya Viswanathan, Mirjana Spasojevic, Janet Go, Karthik Acharya, and Robert Ahonius. 2010. Fear and the city: role of mobile services in harnessing safety and security in urban use contexts. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*. Association for Computing Machinery, New York, NY, USA, 1841–1850. DOI: <https://doi.org/10.1145/1753326.1753602>
- [2] Chaeyoon Yoo and Paul Dourish. 2021. Anshimi: Women's Perceptions of Safety Data and the Efficacy of a Safety Application in Seoul. *Proc. ACM Hum.-Comput.*

- Interact. 5, CSCW1, Article 147 (April 2021), 21 pages. DOI:https://doi.org/10.1145/3449221
- [3] Syed Ishtiaque Ahmed, Steven J. Jackson, Nova Ahmed, Hasan Shahid Ferdous, Md. Rashidujjaman Rifat, A.S.M Rizvi, Shamir Ahmed, and Rifat Sabbir Mansur. 2014. Prohibit: A platform for fighting sexual harassment in urban Bangladesh. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14). Association for Computing Machinery, New York, NY, USA, 2695–2704. DOI:https://doi.org/10.1145/2556288.2557376
 - [4] Mohammed Eunus Ali, Shabnam Basera Rishta, Lazima Ansari, Tanzima Hashem, and Ahamad Imtiaz Khan. 2015. SafeStreet: empowering women against street harassment using a privacy-aware location based application. In Proceedings of the Seventh International Conference on Information and Communication Technologies and Development (ICTD '15). Association for Computing Machinery, New York, NY, USA, Article 24, 1–4. DOI:https://doi.org/10.1145/2737856.2737870
 - [5] Chelsea Young. 2014. HarassMap: Using Crowdsourced Data to Map Sexual Harassment in Egypt. Technology Innovation Management Review. 4(3), 7–13, https://doi.org/10.22215/timreview/770
 - [6] Christine Satchell and Marcus Foth. 2011. Welcome to the jungle: HCI after dark. In CHI '11 Extended Abstracts on Human Factors in Computing Systems (CHI EA '11). Association for Computing Machinery, New York, NY, USA, 753–762. DOI:https://doi.org/10.1145/1979742.1979630
 - [7] Christine Satchell and Marcus Foth. 2010. Fear and danger in nocturnal urban environments. In Proceedings of the 22nd Conference of the Computer-Human Interaction Special Interest Group of Australia on Computer-Human Interaction (OZCHI '10). Association for Computing Machinery, New York, NY, USA, 380–383. DOI:https://doi.org/10.1145/1952222.1952308
 - [8] Manaswi Saha, Michael Saugstad, Hanuma Teja Maddali, Aileen Zeng, Ryan Holland, Steven Bower, Aditya Dash, Sage Chen, Anthony Li, Kotaro Hara, and Jon Froehlich. 2019. Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data At Scale. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19). Association for Computing Machinery, New York, NY, USA, Paper 62, 1–14. DOI:https://doi.org/10.1145/3290605.3300292
 - [9] Carlos Cardonha, Diego Gallo, Priscilla Avegliano, Ricardo Herrmann, Fernando Koch, and Sergio Borger. 2013. A crowdsourcing platform for the construction of accessibility maps. In Proceedings of the 10th International Cross-Disciplinary Conference on Web Accessibility (W4A '13). Association for Computing Machinery, New York, NY, USA, Article 26, 1–4. DOI:https://doi.org/10.1145/2461121.2461129
 - [10] Federica Cena, Amon Rapp, Maurizio Tirassa, Guido Boella, Alessia Calafiore, and Roberto Keller. 2017. Personalized interactive urban maps for autism: enhancing accessibility to urban environments for people with autism spectrum disorder. In Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers (UbiComp '17). Association for Computing Machinery, New York, NY, USA, 9–12. DOI:https://doi.org/10.1145/3123024.3123148
 - [11] Kirsten Rasmus-Gröhn and Charlotte Magnusson. 2014. Finding the way home: supporting wayfinding for older users with memory problems. In Proceedings of the 8th Nordic Conference on Human-Computer Interaction: Fun, Fast, Foundational (NordiCHI '14). Association for Computing Machinery, New York, NY, USA, 247–255. DOI:https://doi.org/10.1145/2639189.2639233
 - [12] Morgan Klaus Scheuerman, Stacy M. Branham, and Foad Hamidi. 2018. Safe Spaces and Safe Places: Unpacking Technology-Mediated Experiences of Safety and Harm with Transgender People. Proc. ACM Hum.-Comput. Interact. 2, CSCW, Article 155 (November 2018), 27 pages. DOI:https://doi.org/10.1145/3274424
 - [13] Denny L. Starks, Tawanna Dillahunt, and Oliver L. Haimson. 2019. Designing Technology to Support Safety for Transgender Women & Non-Binary People of Color. In Companion Publication of the 2019 on Designing Interactive Systems Conference 2019 Companion (DIS '19 Companion). Association for Computing Machinery, New York, NY, USA, 289–294. DOI:https://doi.org/10.1145/3301019.3323898
 - [14] Kimberle Crenshaw. 1991. Mapping the Margins: Intersectionality, Identity Politics, and Violence Against Women of Color. Stanford Law Review. 43(6). Stanford University School of Law, 1241–99. DOI:https://doi.org/10.2307/1229039
 - [15] Ari Schlesinger, W. Keith Edwards, and Rebecca E. Grinter. 2017. Intersectional HCI: Engaging Identity through Gender, Race, and Class. In Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17). Association for Computing Machinery, New York, NY, USA, 5412–5427. DOI:https://doi-org.offcampus.lib.washington.edu/10.1145/3025453.3025766
 - [16] Marisol Wong-Villacres, Arkadeep Kumar, Aditya Vishwanath, Naveena Karusala, Betsy DiSalvo, and Neha Kumar. 2018. Designing for Intersections. In Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18). Association for Computing Machinery, New York, NY, USA, 45–58. DOI:https://doi.org/10.1145/3196709.3196794
 - [17] Rita Kaur Dhamoon. 2011. Considerations on Mainstreaming Intersectionality. Political Research Quarterly, 64(1). Sage Publications, 230–43. DOI:https://doi.org/10.1177/1065912910379227
 - [18] Rita Kaur Dhamoon. 2015. A feminist approach to decolonizing anti-racism: Rethinking transnationalism, intersectionality, and settler colonialism. Feral Feminisms, 4, 20–37
 - [19] Erete, S., Israni, A., and Dillahunt, T. An intersectional approach to designing in the margins. Interactions 25, 3 (Mar.–Apr. 2018), 66–69.
 - [20] Jill Palzkill Woelfer, Amy Iverson, David G. Hendry, Batya Friedman, and Brian T. Gill. 2011. Improving the safety of homeless young people with mobile phones: values, form and function. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11). Association for Computing Machinery, New York, NY, USA, 1707–1716. DOI:https://doi.org/10.1145/1978942.1979191
 - [21] Batya Friedman and David G. Hendry. Value sensitive design: Shaping technology with moral imagination. MIT Press, Cambridge, MA.
 - [22] Batya Friedman. 1996. Value-sensitive design. interactions 3(6) 16–23.
 - [23] Esther Galbrun, Konstantinos Pelechris, Evimaria Terzi. 2016. Urban navigation beyond shortest route: The case of safe paths. Information Systems, Volume 57, 2016, Pages 160–171, ISSN 0306-4379, https://doi.org/10.1016/j.is.2015.10.005.
 - [24] Matthew Garvey, Nilaksh Das, Jiaxing Su, Meghna Natraj, and Bhanu Verma. 2016. PASSAGE: A Travel Safety Assistant with Safe Path Recommendations for Pedestrians. In Companion Publication of the 21st International Conference on Intelligent User Interfaces (IUI '16 Companion). Association for Computing Machinery, New York, NY, USA, 84–87. https://doi.org/10.1145/2876456.2879470
 - [25] Félix Mata, Miguel Torres-Ruiz, Giovanni Guzmán, Rolando Quintero, Roberto Zagal-Flores, Marco Moreno-Ibarra, Eduardo Loza. 2016. A Mobile Information System Based on Crowd-Sensed and Official Crime Data for Finding Safe Routes: A Case Study of Mexico City. Mobile Information Systems, vol. 2016, Article ID 8068209, 11 pages. https://doi.org/10.1155/2016/8068209
 - [26] Sumit Shah, Fenyue Bao, Chang-Tien Lu, and Ing-Ray Chen. 2011. CROWDSAFE: crowd sourcing of crime incidents and safe routing on mobile devices. In Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS '11). Association for Computing Machinery, New York, NY, USA, 521–524. https://doi.org/10.1145/2093973.2094064
 - [27] N. Goel, R. Sharma, N. Nikhil, S. D. Mahanoor and M. Saini. 2017. A Crowd-Sourced Adaptive Safe Navigation for Smart Cities. 2017 IEEE International Symposium on Multimedia (ISM), pp. 382–387, doi: 10.1109/ISM.2017.77.
 - [28] Yuki Matsuda and Ismail Arai. 2014. A safety assessment system for sidewalks at night utilizing smartphones' light sensors. In Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication (UbiComp '14 Adjunct). Association for Computing Machinery, New York, NY, USA, 115–118. DOI:https://doi.org/10.1145/2638728.2638797
 - [29] Daniela K. Rosner, Margaret E. Morris, Ariel Duncan, Sarah E. Fox, Kathi R. Kitner, Ankur Agrawal, and Mei J. Chen. 2016. Designing for Movement in Public Life with Itinerant Probes. In Proceedings of the 2016 ACM Conference on Designing Interactive Systems (DIS '16). Association for Computing Machinery, New York, NY, USA, 1072–1082. DOI:https://doi.org/10.1145/2901790.2901913
 - [30] Asif Iqbal Middy, Sarbani Roy, and Debjani Chattopadhyay. 2021. CityLightSense: A Participatory Sensing-based System for Monitoring and Mapping of Illumination Levels. ACM Trans. Spatial Algorithms Syst. 8, 1, Article 5 (March 2022), 22 pages. DOI:https://doi.org/10.1145/3487364
 - [31] A. Kumar U. and B. Adityan, "A Mobile-based personal safety app to detect well-lit streets: for safe night-time travel," 2020 Third International Conference on Smart. Systems and Inventive Technology (ICSSIT), 2020, pp. 207–214, doi: 10.1109/ICSSIT48917.2020.9214128.
 - [32] Kayur Patel, Mike Y. Chen, Ian Smith, and James A. Landay. 2006. Personalizing routes. In Proceedings of the 19th annual ACM symposium on User interface software and technology (UIST' 06). Association for Computing Machinery, New York, NY, USA, 187–190. https://doi.org/10.1145/1166253.1166282
 - [33] Panurat Sutigoolabud, Peerawit Naprae, and Santi Phithakkitnukoon. 2019. Safe street rangers: crowdsourcing approach for monitoring and reporting street safety. In Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers (UbiComp/ISWC '19 Adjunct). Association for Computing Machinery, New York, NY, USA, 1014–1021. DOI:https://doi.org/10.1145/3341162.3349317
 - [34] Ihudiya Finda Ogbonnaya-Ogburu, Angela D R Smith, Alexandra To, and Kentaro Toyama. 2020. Critical race theory for HCI. Proceedings of the 2020 CHI conference on human factors in computing systems, (Apr. 2020), 1–16. DOI: https://doi.org/10.1145/3313831.3376392
 - [35] Robert Stuart Weiss. 1994. Learning from Strangers: The Art and Method of Qualitative Interview Studies. Simon and Schuster, New York, NY.
 - [36] John W. Creswell and Cheryl N. Poth. 2016. Qualitative inquiry and research design: Choosing among five approaches. Sage Publications, Thousand Oaks, CA.
 - [37] Virginia Braun and Victoria Clarke. 2012. Thematic analysis. APA Handbook of Research Methods in Psychology, Vol 2: Research Designs: Quantitative, Qualitative, Neuropsychological, and Biological, 57–71. DOI: https://doi.org/10.1037/13620-004
 - [38] Sarah E. Fox, Rafael ML Silva, and Daniela K. Rosner. 2018. Beyond the prototype: Maintenance, collective responsibility, and public IoT. Proceedings of the 2018 Designing Interactive Systems Conference, (Jun. 2018), 21–32. DOI: https://doi.org/10.1145/3201019.3201021

- org/10.1145/3196709.3196710
- [39] Russell Dicker. 2021. 3 new ways to navigate more sustainably with Maps. Retrieved April 21, 2022 from <https://blog.google/products/maps/3-new-ways-navigate-more-sustainably-maps/>
 - [40] Traunmueller, M., Marshall, P., Capra, L. (2015). Crowdsourcing Safety Perceptions of People: Opportunities and Limitations. In: Liu, TY., Scollon, C., Zhu, W. (eds) Social Informatics. SocInfo 2015. Lecture Notes in Computer Science(), vol 9471. Springer, Cham. https://doi.org/10.1007/978-3-319-27433-1_9
 - [41] Solymosi, R., Buil-Gil, D., Vozmediano, L., & Guedes, I. S. (2021). Towards a Place-based Measure of Fear of Crime: A Systematic Review of App-based and Crowdsourcing Approaches. *Environment and Behavior*, 53(9), 1013–1044. <https://doi.org/10.1177/0013916520947114>
 - [42] Harald Holone, Gunnar Misund, Håkon Tolsby, and Steinar Kristoffersen. 2008. Aspects of personal navigation with collaborative user feedback. In *Proceedings of the 5th Nordic conference on Human-computer interaction: building bridges (NordiCHI '08)*. Association for Computing Machinery, New York, NY, USA, 182–191. <https://doi.org/10.1145/1463160.1463180>
 - [43] Fileborn, B., & O'Neill, T. (2023). From “Ghettoization” to a Field of Its Own: A Comprehensive Review of Street Harassment Research. *Trauma, Violence, & Abuse*, 24(1), 125–138. <https://doi.org/10.1177/15248380211021608-/bib>>
 - [44] Mark S. Ackerman. 2000. The intellectual challenge of CSCW: the gap between social requirements and technical feasibility. *Hum.-Comput. Interact.* 15, 2 (September 2000), 179–203. https://doi.org/10.1207/S15327051HCI1523_5

APPENDIX

Table 1: Participant Demographics

Participant Number	How do you identify your gender?	How do you describe your race?	What is your age group?	What other modes of transportation (besides walking) do you use at least once a week?
P1	Woman	Irish	65+	Bus
P2	Man	Asian	18-65	Light rail
P3	Woman	White	65+	Bus, Driving
P4	Man	American Indian or Alaska Native, White	18-65	Bus, light rail
P5	Woman	White	65+	Bus
P6	Woman	Asian	18-65	Driving
P7	Woman	Black or African American	18-65	Bus, Light Rail, Driving
P8	Man	Black or African American	18-65	Bus
P9	Woman	Asian	18-65	Bus, Light rail
P10	Woman	Asian	18-65	Driving
P11	Non-binary	Asian	18-65	Bus, Light rail
P12	Woman	Asian	18-65	Bus, Light rail